Ultrasound Medical Imager

Final Dimensions: 10*14 cm^2

3D Model

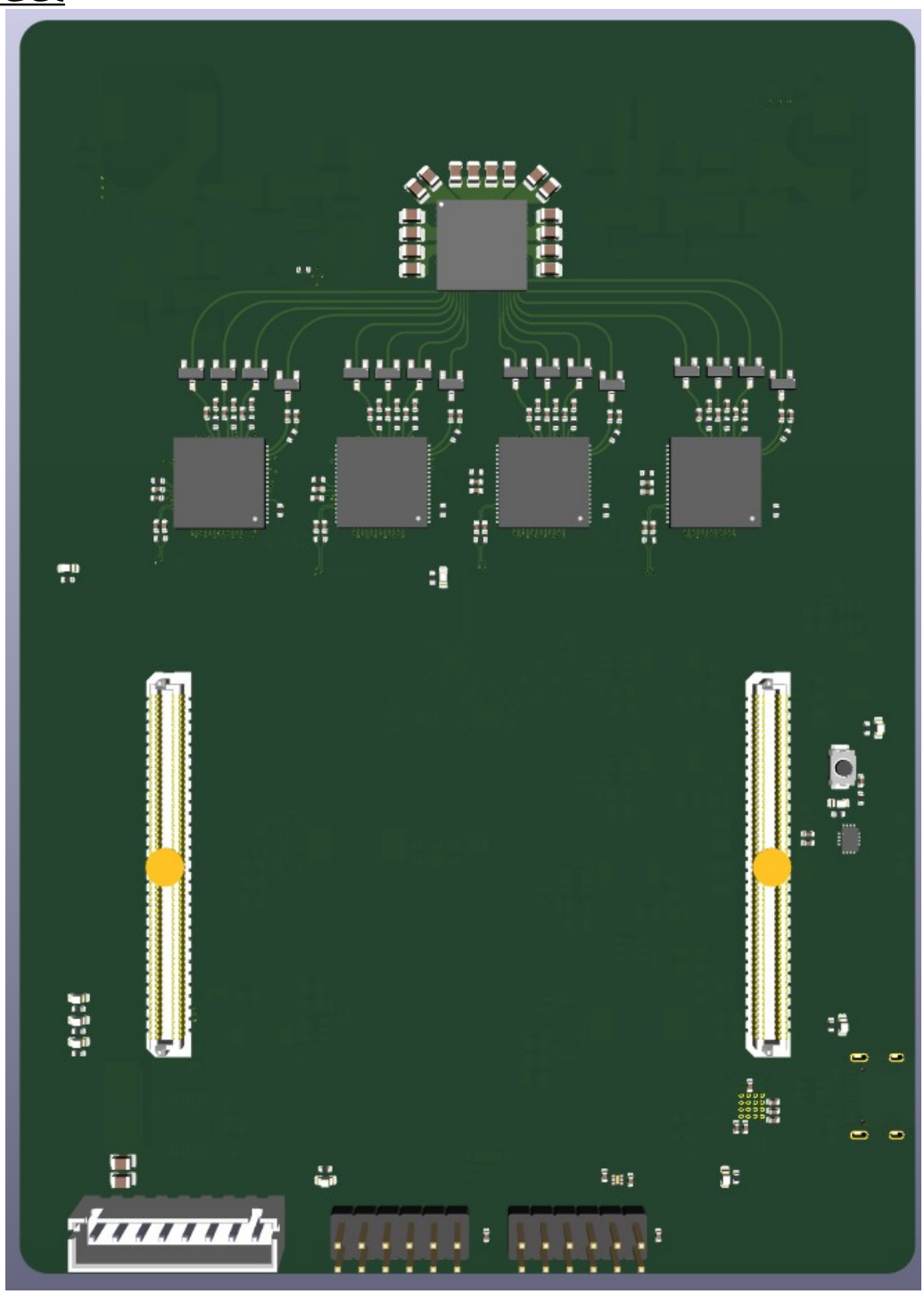


Figure 1: Top view of the PCB.

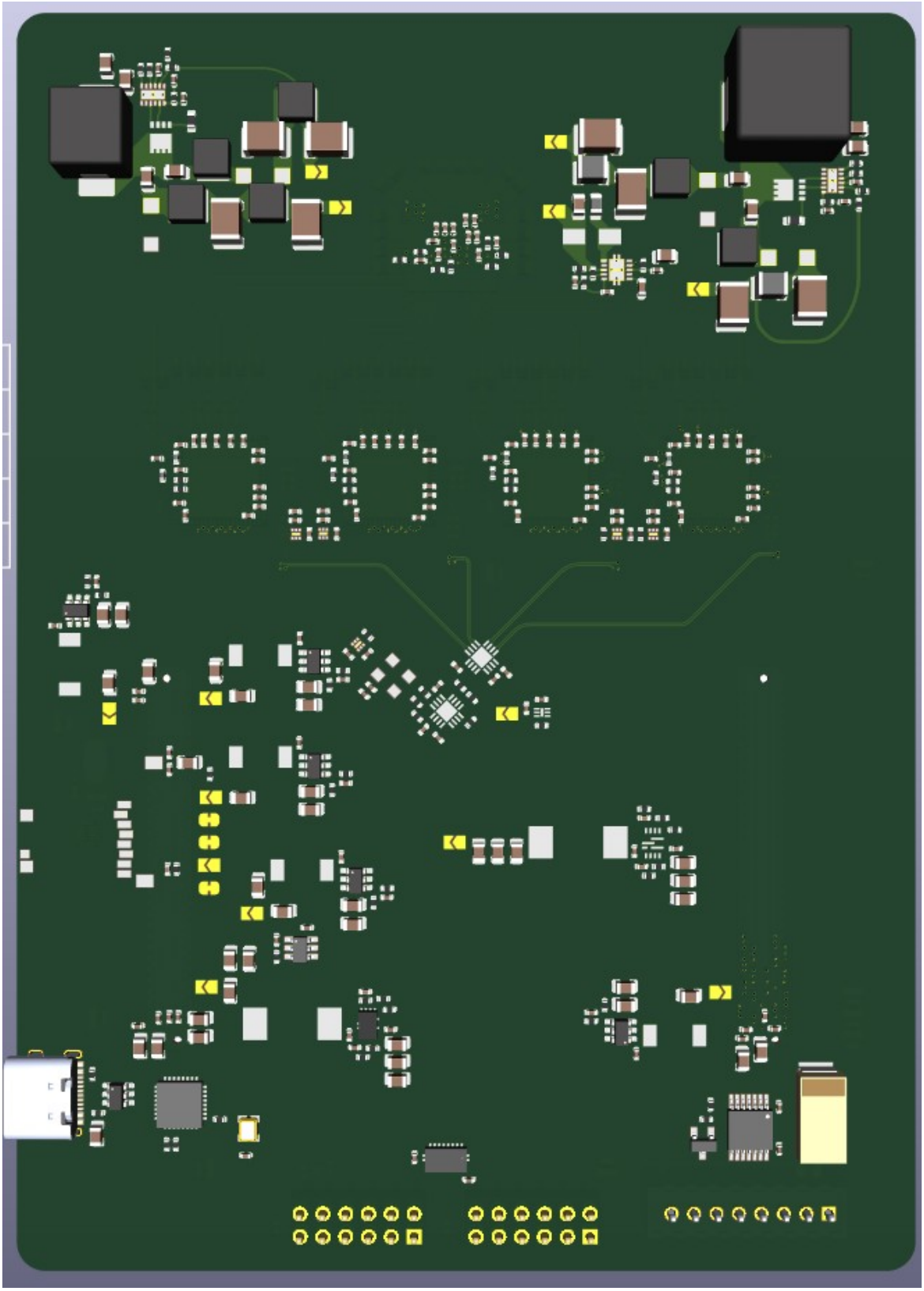
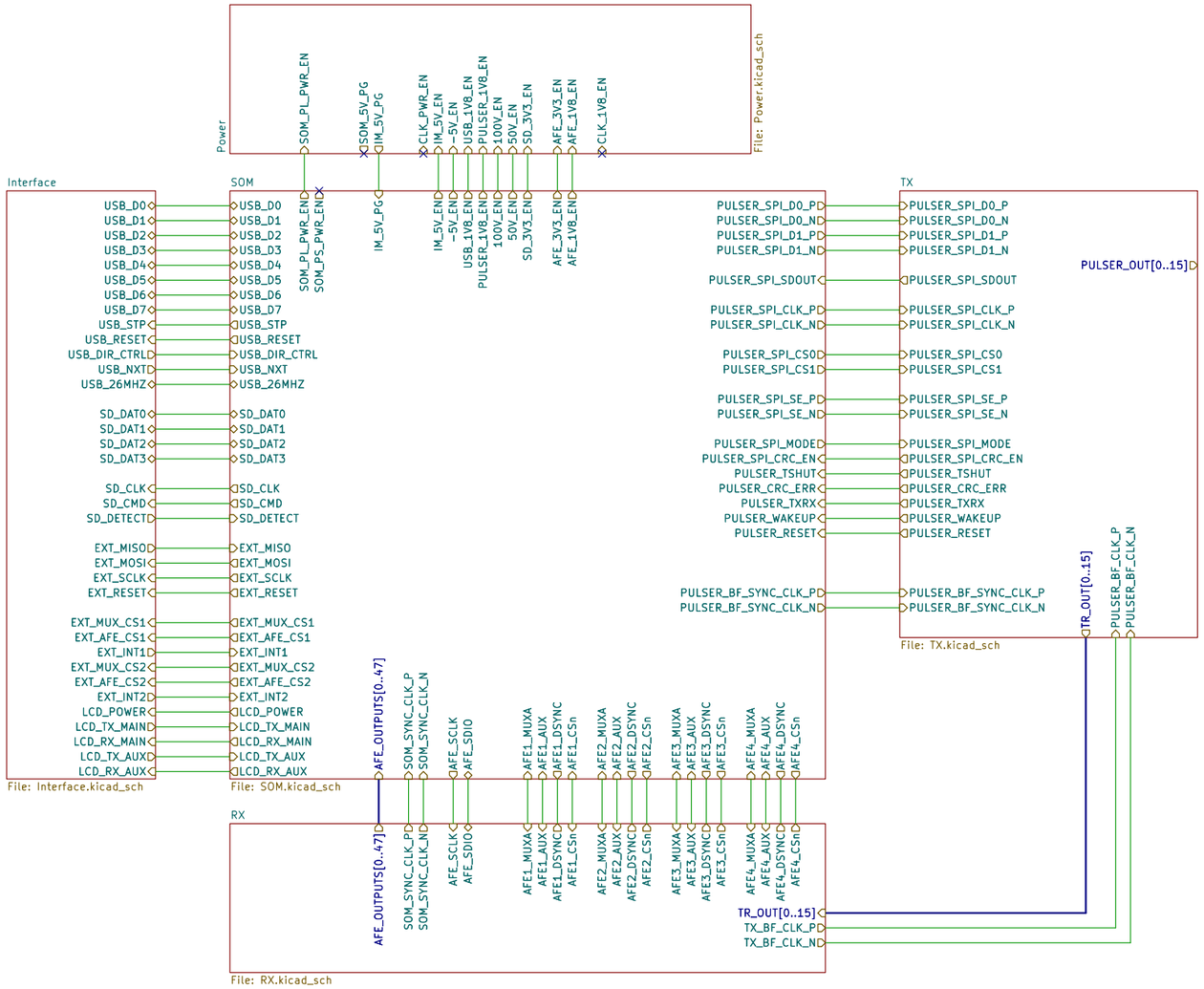
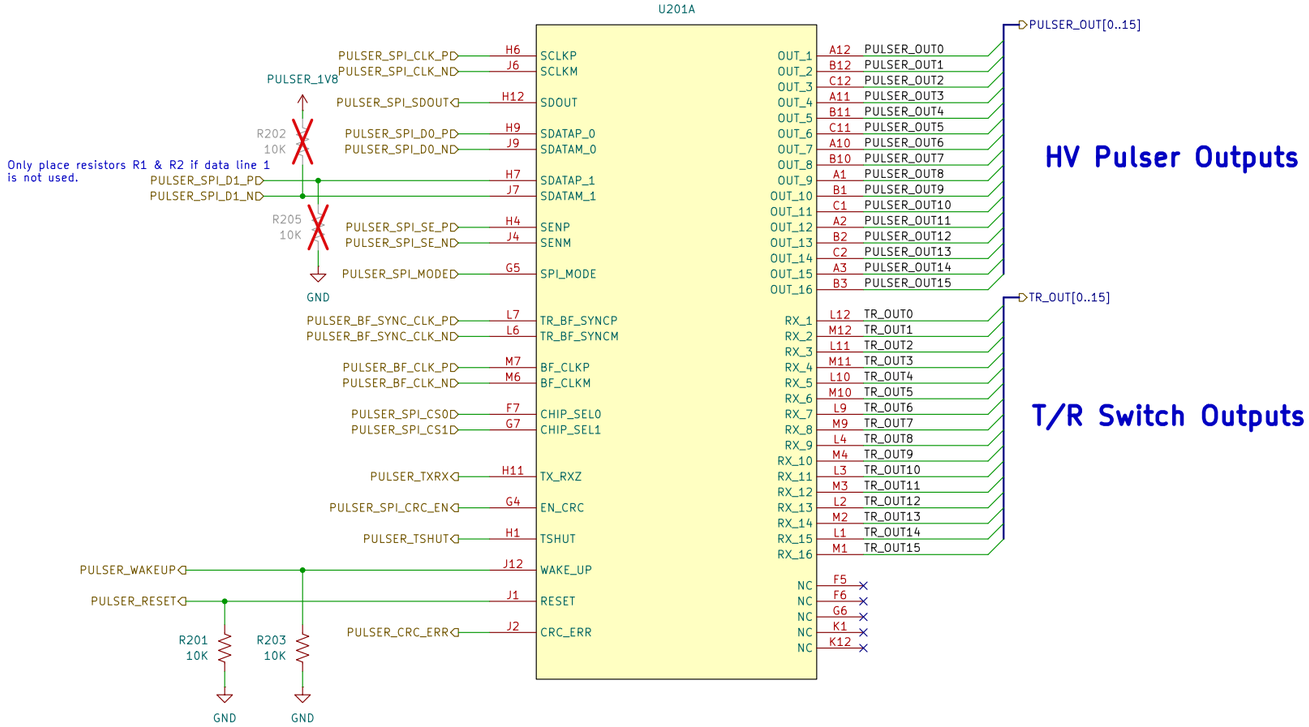


Figure 2: Bottom view of the PCB.

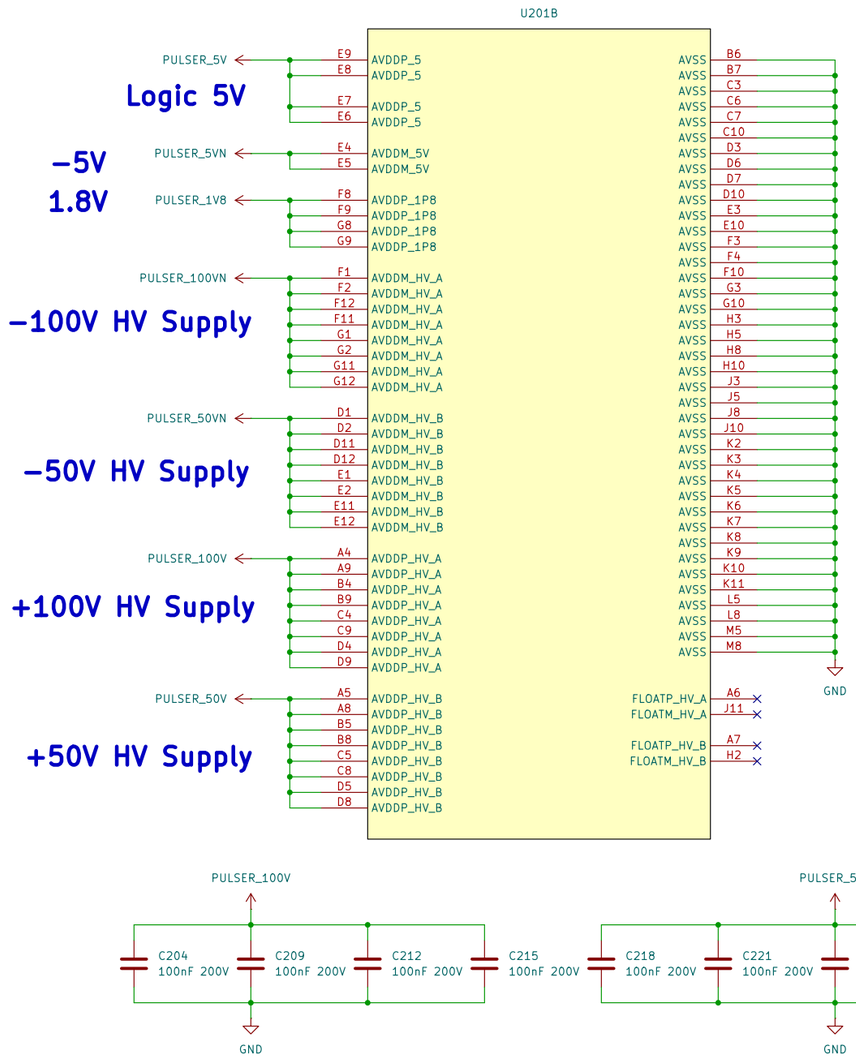


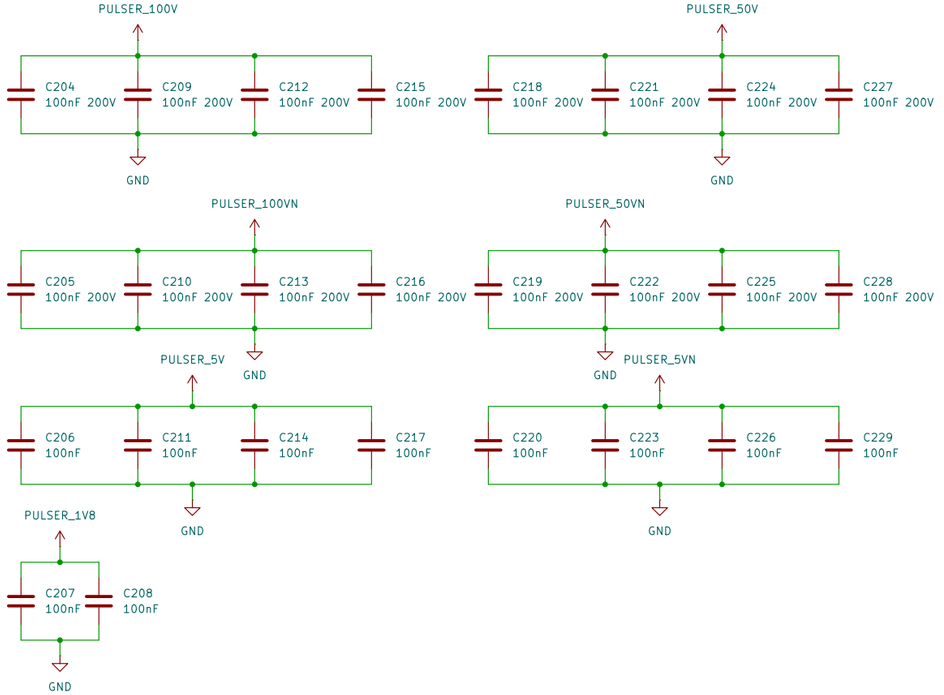
Power Adapter	12-24V (5A)	SOM Buck Converter	5V 4A (1.6A @ 12V)	Main SOM Supply (4A)	
		Intermediate Voltage Buck Converter	5V 2A (0.8A @ 12V)	Pulser Main Voltage (5V, 0.016A)	
				Pulser Negative Voltage (-5V, 0.023A)	
				AFE 1 & 2 Main Supply (3.3V, 0.26A)	
				AFE 1 & 2 Logic Supply (1.8V, 0.3A)	
				AFE 3 & 4 Main Supply (3.3V, 0.26A)	
				AFE 3 & 4 Logic Supply (1.8V, 0.3A)	
				Clock LDO Regulator 1	3.3V 0.1A
				Clock LDO Regulator 2	1.8V 0.1A
				External Interfaces LDO Regulator	3.3V ?A Up to 2.5A
		HDA Buck Converter	3.3V 1A (0.3A @ 12V)		
		HDB & HDC Buck Converter	3.3V 1A (0.3A @ 12V)		
		HPA Buck Converter	1.8V 1A (0.3A @ 12V)		
		USB & Pulser Buck Converter	1.8V 0.064A		
		SD Card Buck Converter	3.3V 1.2A (0.35A @ 12V)		
		+/-100V HV Converter	+/-100V 0.0005A (0.004A @ 12V)		
		+/-50V HV Converter	+/-50V 0.3A (1.2A @ 12V)		

TODO:
- **DETAIL THE INDUCTORS' FOOTPRINTS AT THE END**
- **ENSURE STRAPPING RESISTORS ARE AT ALL CS/EN PINS**
- **Add LEDs at all unconnected PG signals**


U201A
PULSER_SPI_CLK_PD
PULSER_SPI_CLK_ND
PULSER_1V8
PULSER_SPI_SDOUT
R202
10K
PULSER_SPI_DO_PD
PULSER_SPI_DO_ND
Only place resistors R1 & R2 if data line 1 is not used.
PULSER_SPI_D1_PD
PULSER_SPI_D1_ND
R205
10K
PULSER_SPI_SE_PD
PULSER_SPI_SE_ND
GND
PULSER_SPI_MODE
PULSER_BF_SYNC_CLK_PD
PULSER_BF_SYNC_CLK_ND
PULSER_BF_CLK_PD
PULSER_BF_CLK_ND
PULSER_SPI_CS0
PULSER_SPI_CS1
PULSER_TXRX
PULSER_SPI_CRC_EN
PULSER_TSHUT
PULSER_WAKEUP
PULSER_RESET
R201
10K
R203
10K
PULSER_CRC_ERR
GND
GND
H6
J6
H12
H9
J9
H7
J7
H4
J4
G5
L7
L6
M7
M6
F7
G7
H11
G4
H1
J12
J1
J2
SCLKP
SCLKM
SDOUT
SDATAP_0
SDATAM_0
SDATAP_1
SDATAM_1
SENP
SENM
SPI_MODE
TR_BF_SYNCP
TR_BF_SYNCM
BF_CLKP
BF_CLKM
CHIP_SEL0
CHIP_SEL1
TX_RXZ
EN_CRC
TSHUT
WAKE_UP
RESET
CRC_ERR
OUT_1
OUT_2
OUT_3
OUT_4
OUT_5
OUT_6
OUT_7
OUT_8
OUT_9
OUT_10
OUT_11
OUT_12
OUT_13
OUT_14
OUT_15
OUT_16
A12 PULSER_OUT0
B12 PULSER_OUT1
C12 PULSER_OUT2
A11 PULSER_OUT3
B11 PULSER_OUT4
C11 PULSER_OUT5
A10 PULSER_OUT6
B10 PULSER_OUT7
A1 PULSER_OUT8
B1 PULSER_OUT9
C1 PULSER_OUT10
A2 PULSER_OUT11
B2 PULSER_OUT12
C2 PULSER_OUT13
A3 PULSER_OUT14
B3 PULSER_OUT15
PULSER_OUT[0..15]
HV Pulser Outputs
RX_1
RX_2
RX_3
RX_4
RX_5
RX_6
RX_7
RX_8
RX_9
RX_10
RX_11
RX_12
RX_13
RX_14
RX_15
RX_16
L12 TR_OUT0
M12 TR_OUT1
L11 TR_OUT2
M11 TR_OUT3
L10 TR_OUT4
M10 TR_OUT5
L9 TR_OUT6
M9 TR_OUT7
L4 TR_OUT8
M4 TR_OUT9
L3 TR_OUT10
M3 TR_OUT11
L2 TR_OUT12
M2 TR_OUT13
L1 TR_OUT14
M1 TR_OUT15
TR_OUT[0..15]
T/R Switch Outputs
NC
NC
NC
NC
NC
F5
F6
G6
K1
K12


Output:
Indicates whether the pulser is in transmit or receiver mode.
PULSER_TXRX
R204
10K
C201
1nF
GND
PULSER_TSHUT
R206
10K
C202
1nF
GND
Output:
Thermal shutdown flag.
PULSER_CRC_ERR
R207
10K
C203
1nF
GND
Output:
SPI error flag (CRC error).


U201B
PULSER_5V
Logic 5V
PULSER_5VN
−5V
1.8V
PULSER_1V8
PULSER_100VN
−100V HV Supply
PULSER_50VN
−50V HV Supply
PULSER_100V
+100V HV Supply
PULSER_50V
+50V HV Supply
E9
E8
E7
E6
E4
E5
F8
F9
G8
G9
F1
F2
F12
F11
G1
G2
G11
G12
D1
D2
D11
D12
E1
E2
E11
E12
A4
A9
B4
B9
C4
C9
D4
D9
A5
A8
B5
B8
C5
C8
D5
D8
AVDDP_5
AVDDM_5V
AVDDP_1P8
AVDDM_HV_A
AVDDM_HV_B
AVDDP_HV_A
AVDDP_HV_B
B6
B7
C3
C6
C7
C10
D3
D6
D7
D10
E3
E10
F3
F4
F10
G3
G10
H3
H5
H8
H10
J3
J5
J8
J10
K2
K3
K4
K5
K6
K7
K8
K9
K10
K11
L5
L8
M5
M8
AVSS
GND
A6
J11
A7
H2
FLOATP_HV_A
FLOATM_HV_A
FLOATP_HV_B
FLOATM_HV_B


PULSER_100V
C204
100nF 200V
C209
100nF 200V
C212
100nF 200V
C215
100nF 200V
GND
PULSER_50V
C218
100nF 200V
C221
100nF 200V
C224
100nF 200V
C227
100nF 200V
GND
PULSER_100VN
C205
100nF 200V
C210
100nF 200V
C213
100nF 200V
C216
100nF 200V
GND
PULSER_50VN
C219
100nF 200V
C222
100nF 200V
C225
100nF 200V
C228
100nF 200V
GND
PULSER_5V
C206
100nF
C211
100nF
C214
100nF
C217
100nF
GND
PULSER_5VN
C220
100nF
C223
100nF
C226
100nF
C229
100nF
GND
PULSER_1V8
C207
100nF
C208
100nF
GND

Fanout Buffer 1

Feeds:
SOM sync signal
Pulser beamformer sync signal
AFE fanout buffer –>

Table 8-1. Input Selection Table

IN_SEL	ACTIVE CLOCK INPUT
0	IN0_P, IN0_N
1	IN1_P, IN1_N
Open	None (1)

72 MHz Crystal

Y301
ASFLMB-72.000MHZ-LY-T

U301
LMK1D1204RGTR

R303 50R
R301 1K
R302 1K
R304 100R
R305 100R
R306 100R

Bias for common mode voltage (0.9v)

It was recommended by the datasheet to terminate used & unused channels.

Fanout Buffer 2

Feeds:
4 * AFE

U302
LMK1D1204RGTR

Terminated at AFE side

C301 10uF
C302 100nF
C305 100nF
C309 10uF
C314 100nF
C323 10uF
C324 100nF
C327 100nF

C303 1uF
C306 1uF
C310 1uF
C312 1uF
C315 1uF
C317 1uF
C319 1uF
C321 1uF

AFE1
File: RX_LANE.kicad_sch

AFE2
File: RX_LANE.kicad_sch

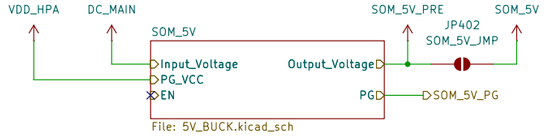
AFE3
File: RX_LANE.kicad_sch

AFE4
File: RX_LANE.kicad_sch

TR_OUT[0..15]
AFE_OUTPUTS[0..47]

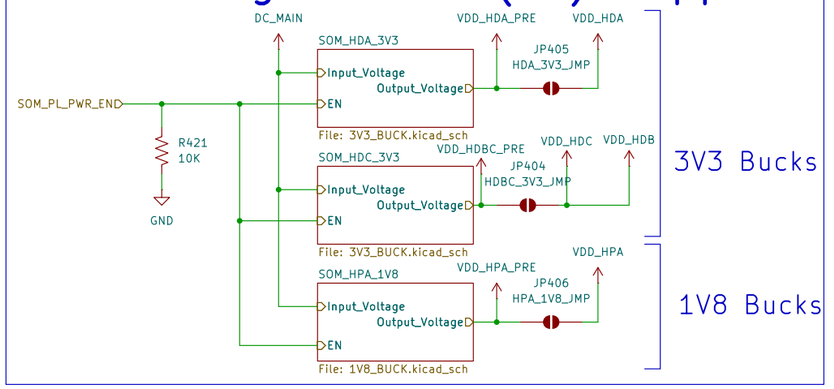
AFE IO are 3V3

Main SOM 5V Supply

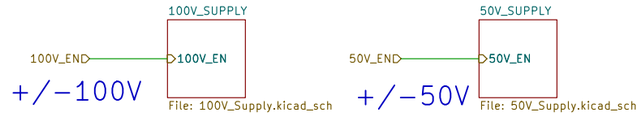


Power Rail Name	Supported Voltage Range	Maximum Current	Description
V_{CC_SOM}	5V (4.75V – 5.25V) 50 mV p-p maximum noise	4A	Main power input to the SOM. Supplies power to on-board power regulators. Total of 4A across connectors. SOM will consume a maximum of 3A.
V_{CC_BATT}	1.20 – 1.50V	150 uA – 300 uA	Nominal battery input for the RTC
V_{CCO_HPA}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 66
V_{CCO_HPB}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 65
V_{CCO_HPC}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 64
V_{CCO_HDA}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 45
V_{CCO_HDB}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 43
V_{CCO_HDC}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 44

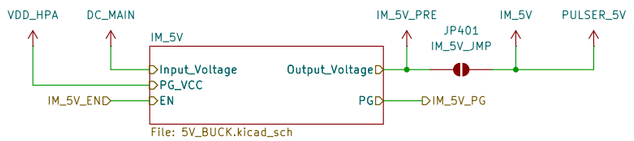
Processing Section (PL) Supplies



HV Supplies (+/- 50/100V)



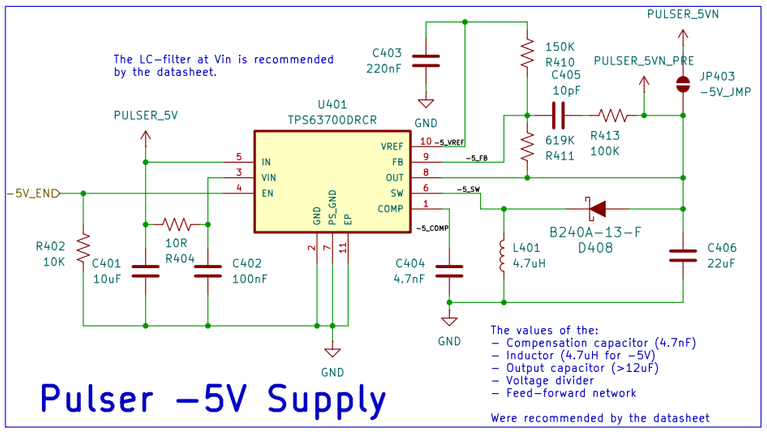
Pulser + Intermediate 5V Supply

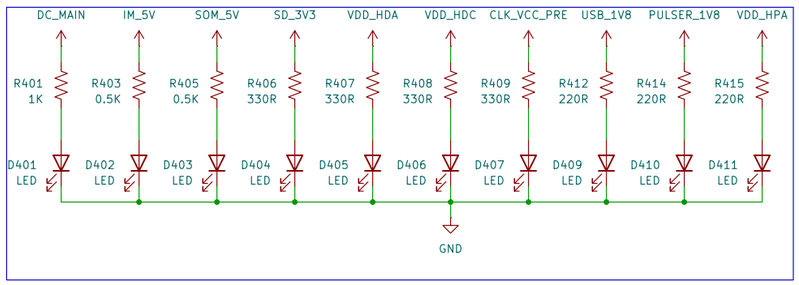


I_{WDD18}	Full channel mode	130	mA
I_{WDD33}	Full channel mode	130	mA
I_{VDD18}		22	mA
I_{VDD33}		2	mA

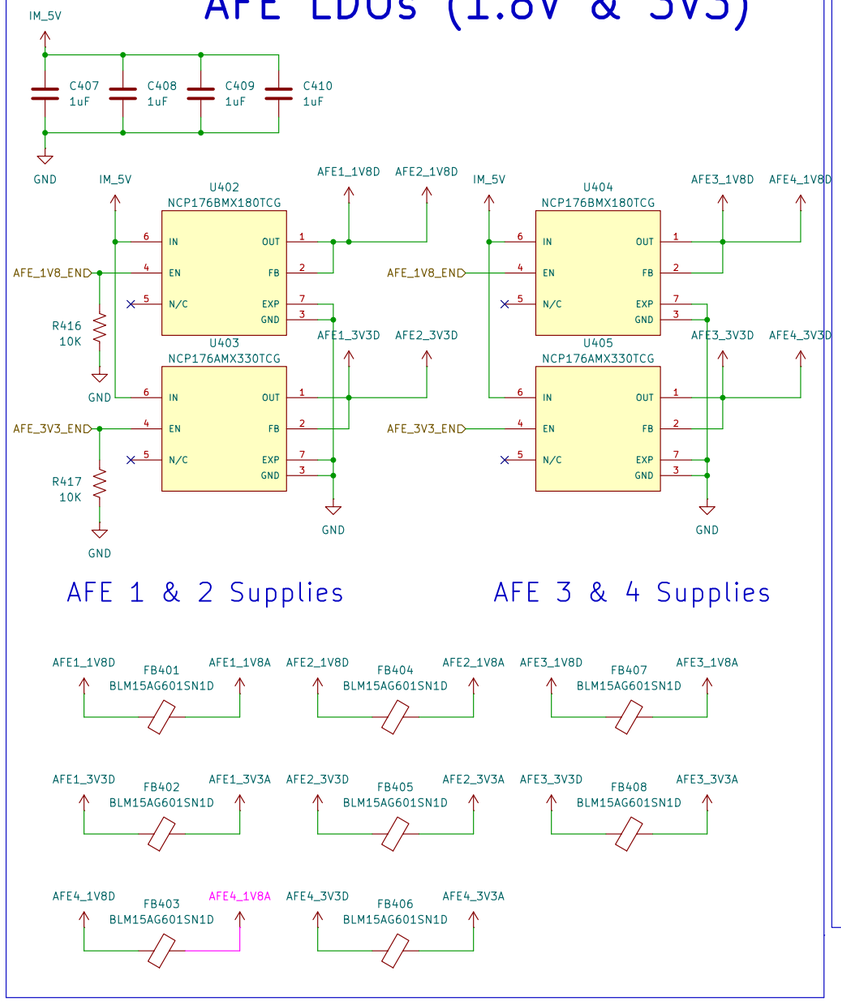
^ x4 = 1.2A

Pulser −5V Supply

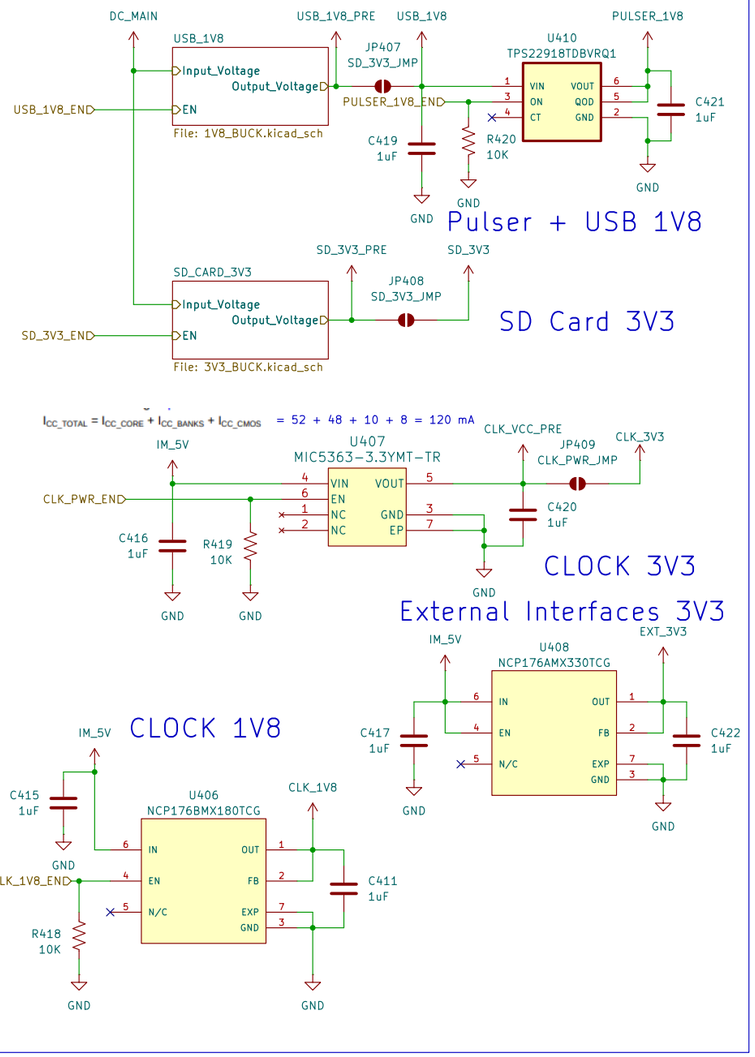




AFE LDOs (1.8V & 3V3)



Miscellaneous Supplies



VCC_BATT	PS BBRAM and real-time clock (RTC) supply voltage, requires external battery. Connect to GND when battery is not used.

J501A
REF-226081-01

J501B
REF-226081-01

J502A
REF-226081-01

J502B
REF-226081-01

SOM240_1

SOM240_2

HPA
HDA
HDB
HDC

PULSER_SPI_D0_P
PULSER_SPI_D0_N
PULSER_SPI_CLK_P
PULSER_SPI_CLK_N
PULSER_SPI_D1_P
PULSER_SPI_D1_N
PULSER_SPI_SE_P
PULSER_SPI_SE_N
PULSER_SPI_CS0
PULSER_SPI_CS1
PULSER_BF_SYNC_CLK_P
PULSER_BF_SYNC_CLK_N
PULSER_TXRX
PULSER_SPI_CRC_EN
PULSER_SPI_SDOUT
PULSER_SPI_MODE
PULSER_1V8_EN
PULSER_TSHUT
PULSER_WAKEUP
PULSER_RESET
PULSER_CRC_ERR

AFE1_DSYNC
AFE1_CSn
AFE1_MUXAD
AFE1_AUX
AFE2_DSYNC
AFE2_CSn
AFE2_MUXA
AFE2_AUX
AFE3_DSYNC
AFE3_CSn
AFE3_MUXAD
AFE3_AUX
AFE4_DSYNC
AFE4_CSn
AFE4_MUXAD
AFE_SCLK
AFE4_AUX
AFE_SDIO
SD_DETECT
EXT_INT1
EXT_INT2

SOM_PS_PWR_EN
SOM_PL_PWR_EN
JTAG_TMS_C2M
JTAG_TDO_M2C
JTAG_TDI_C2M
JTAG_TCK_C2M
BOOT_MODE0
BOOT_MODE1
BOOT_MODE2
BOOT_MODE3

50V_EN
100V_EN
SD_3V3_EN
USB_D5
USB_D6
USB_D7
AFE_1V8_EN
AFE_3V3_EN
USB_1V8_EN

LCD_POWER
LCD_TX_MAIN
LCD_RX_MAIN

USB_STP
USB_D3
USB_D4
EXT_MISO
EXT_MOSI
EXT_SCLK

SD_DAT0
SD_DAT1
SD_DAT2
SD_DAT3
SD_CLK
SD_CMD
USB_RESET
USB_DIR_CTRL
USB_D2
EXT_RESET
EXT_AFE_CS1
EXT_AFE_CS2
IM_5V_EN
-5V_EN

USB_NXT
USB_D0
USB_D1
USB_26MHZ
EXT_MUX_CS1
EXT_MUX_CS2

SOM_SYNC_CLK_PD
SOM_SYNC_CLK_ND

R501
10K
VCC_SOM
PWROFF_ C2M_L

VDD_HPA
VDD_HDA
VDD_HPB
VDD_HDB
VDD_HPC
VDD_HDC
SOM_5V
GND

AFE_OUTPUTS[0..47]
AFE_OUTPUTS0
AFE_OUTPUTS1
AFE_OUTPUTS2
AFE_OUTPUTS3
AFE_OUTPUTS4
AFE_OUTPUTS5
AFE_OUTPUTS6
AFE_OUTPUTS7
AFE_OUTPUTS8
AFE_OUTPUTS9
AFE_OUTPUTS10
AFE_OUTPUTS11
AFE_OUTPUTS12
AFE_OUTPUTS13
AFE_OUTPUTS14
AFE_OUTPUTS15
AFE_OUTPUTS16
AFE_OUTPUTS17
AFE_OUTPUTS18
AFE_OUTPUTS19
AFE_OUTPUTS20
AFE_OUTPUTS21
AFE_OUTPUTS22
AFE_OUTPUTS23
AFE_OUTPUTS24
AFE_OUTPUTS25
AFE_OUTPUTS26
AFE_OUTPUTS27
AFE_OUTPUTS28
AFE_OUTPUTS29
AFE_OUTPUTS30
AFE_OUTPUTS31
AFE_OUTPUTS32
AFE_OUTPUTS33
AFE_OUTPUTS34
AFE_OUTPUTS35
AFE_OUTPUTS36
AFE_OUTPUTS37
AFE_OUTPUTS38
AFE_OUTPUTS39
AFE_OUTPUTS40
AFE_OUTPUTS41
AFE_OUTPUTS42
AFE_OUTPUTS43
AFE_OUTPUTS44
AFE_OUTPUTS45
AFE_OUTPUTS46
AFE_OUTPUTS47

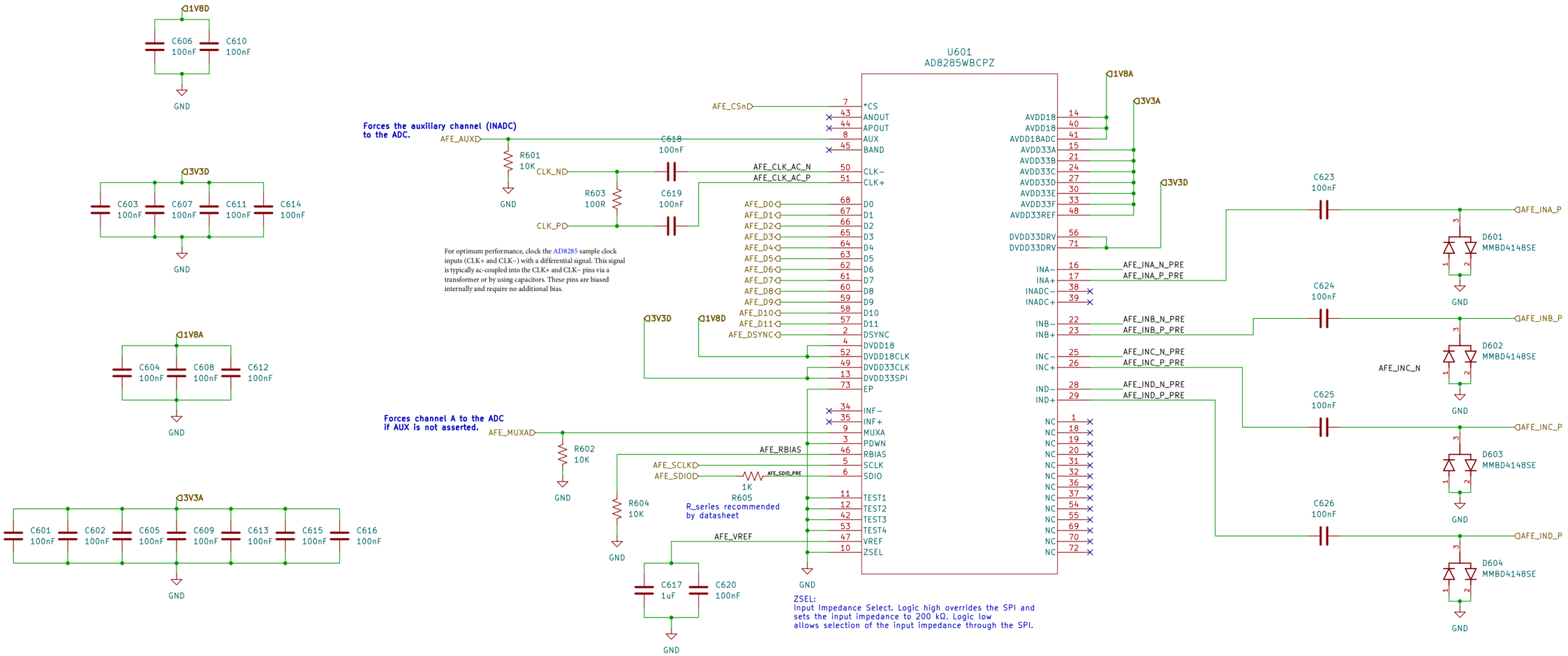
IM_5V_PGD

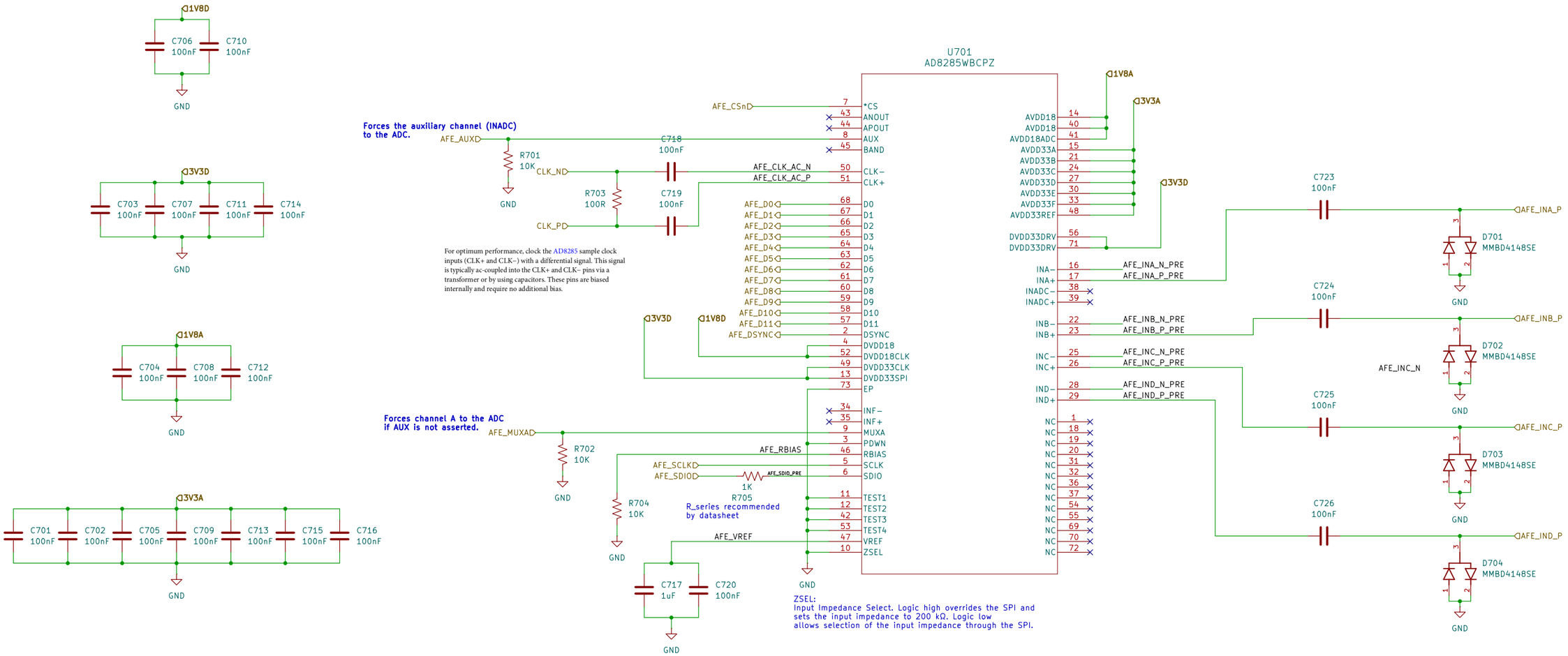
BOOT_MODE0 BOOT_MODE1 BOOT_MODE2 BOOT_MODE3
JP501 BM0
JP502 BM1
JP503 BM2
JP504 BM3

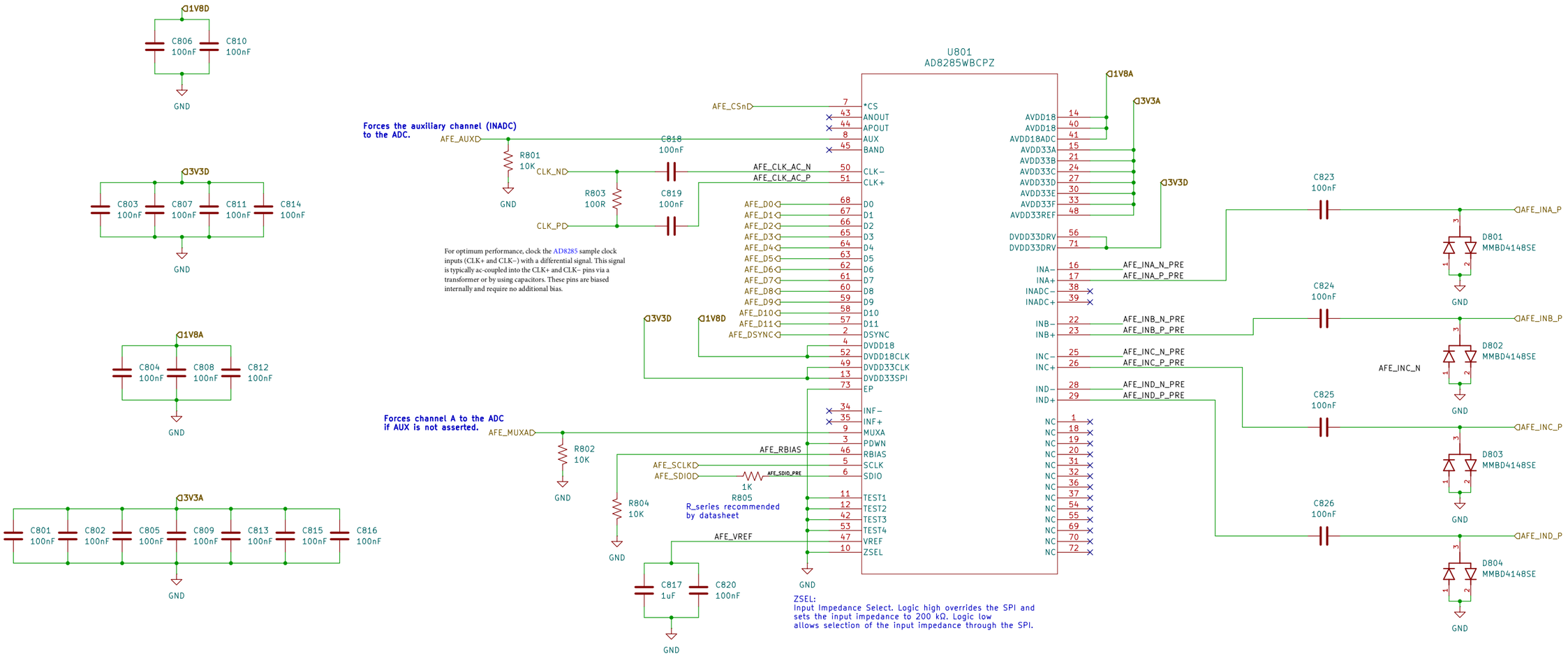
Boot Mode	PS_Mode Pins[3:0]	Physical Pin Location
Quad-SPI (32 bit)	0010	MIO[5:0]
eMMC	0110	MIO[22:13]

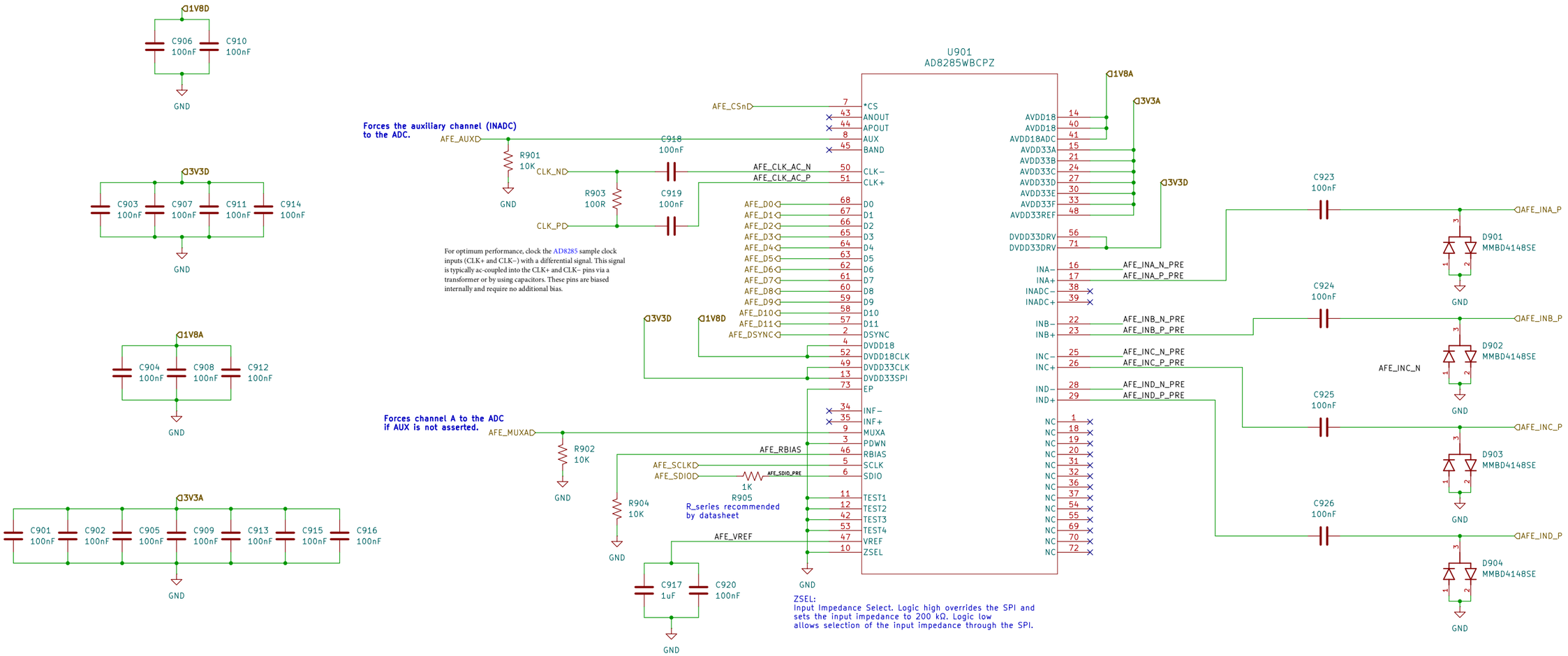
PWROFF_ C2M_L
SW501 RESET
C512 100nF
Reset Button

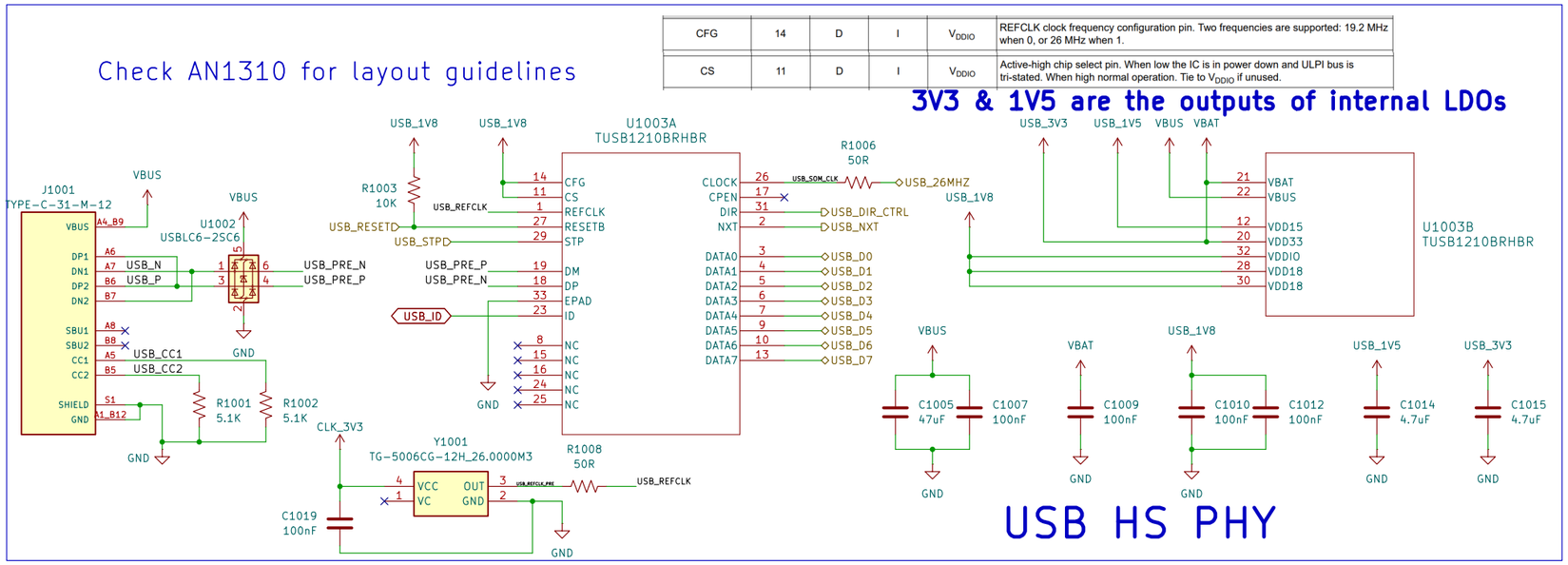
C501 100nF
C502 1uF
C503 47uF
C504 47uF
C505 100nF
C506 1uF
C507 47uF
C508 47uF
C509 4.7uF
C510 100nF
C511 100nF
C513 4.7uF
C514 100nF
C515 100nF
C516 4.7uF
C517 100nF
C518 100nF
C519 4.7uF
C520 100nF
C521 100nF


1V8D
C606 100nF
C610 100nF
GND
3V3D
C603 100nF
C607 100nF
C611 100nF
C614 100nF
GND
1V8A
C604 100nF
C608 100nF
C612 100nF
GND
3V3A
C601 100nF
C602 100nF
C605 100nF
C609 100nF
C613 100nF
C615 100nF
C616 100nF
GND
U601
AD8285WBCPZ
Forces the auxiliary channel (INADC) to the ADC.
AFE_AUXD
R601 10K
GND
C618 100nF
CLK_ND
R603 100R
C619 100nF
CLK_PD
AFE_CSn
AFE_CLK_AC_N
AFE_CLK_AC_P
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
Forces channel A to the ADC if AUX is not asserted.
AFE_MUXAD
R602 10K
GND
AFE_RBIAS
AFE_SCLKD
AFE_SDIOD
AFE_SDIO_PRE
1K
R605
R_series recommended by datasheet
R604 10K
GND
AFE_VREF
C617 1uF
C620 100nF
GND
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8D
3V3D
1V8A
3V3A
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C623 100nF
C624 100nF
C625 100nF
C626 100nF
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
D601 MMBD4148SE
D602 MMBD4148SE
D603 MMBD4148SE
D604 MMBD4148SE
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R606 8R
R607 8R
R608 8R
R609 8R
Change value according to pulser T/R switch out impedance
C621 100nF
C622 100nF
C627 100nF
C628 100nF
GND

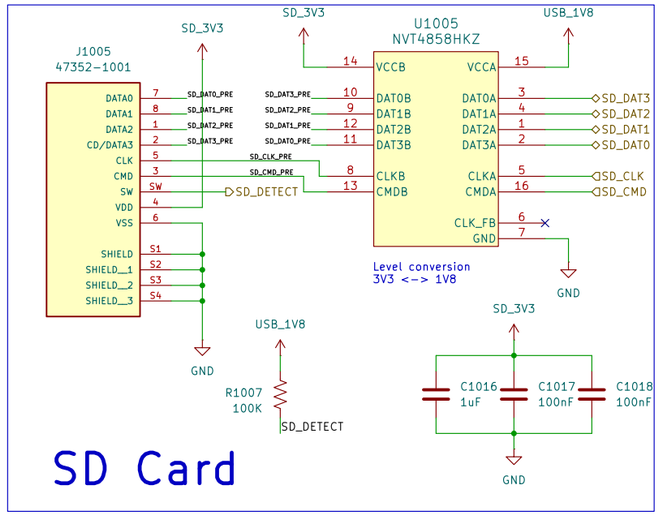

1V8D
C706 100nF
C710 100nF
GND
3V3D
C703 100nF
C707 100nF
C711 100nF
C714 100nF
GND
1V8A
C704 100nF
C708 100nF
C712 100nF
GND
3V3A
C701 100nF
C702 100nF
C705 100nF
C709 100nF
C713 100nF
C715 100nF
C716 100nF
GND
U701
AD8285WBCPZ
Forces the auxiliary channel (INADC) to the ADC.
AFE_AUX
R701 10K
GND
C718 100nF
CLK_N
R703 100R
C719 100nF
CLK_P
AFE_CLK_AC_N
AFE_CLK_AC_P
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
AFE_CSn
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
3V3D
1V8D
Forces channel A to the ADC if AUX is not asserted.
AFE_MUXA
R702 10K
GND
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R705
R_series recommended by datasheet
R704 10K
GND
AFE_VREF
C717 1uF
C720 100nF
GND
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8A
3V3A
3V3D
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
NC
C723 100nF
C724 100nF
C725 100nF
C726 100nF
D701 MMBD4148SE
D702 MMBD4148SE
D703 MMBD4148SE
D704 MMBD4148SE
AFE_INA_P
AFE_INB_P
AFE_INC_N
AFE_INC_P
AFE_IND_P
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R706 8R
R707 8R
R708 8R
R709 8R
Change value according to pulser T/R switch out impedance
C721 100nF
C722 100nF
C727 100nF
C728 100nF
GND

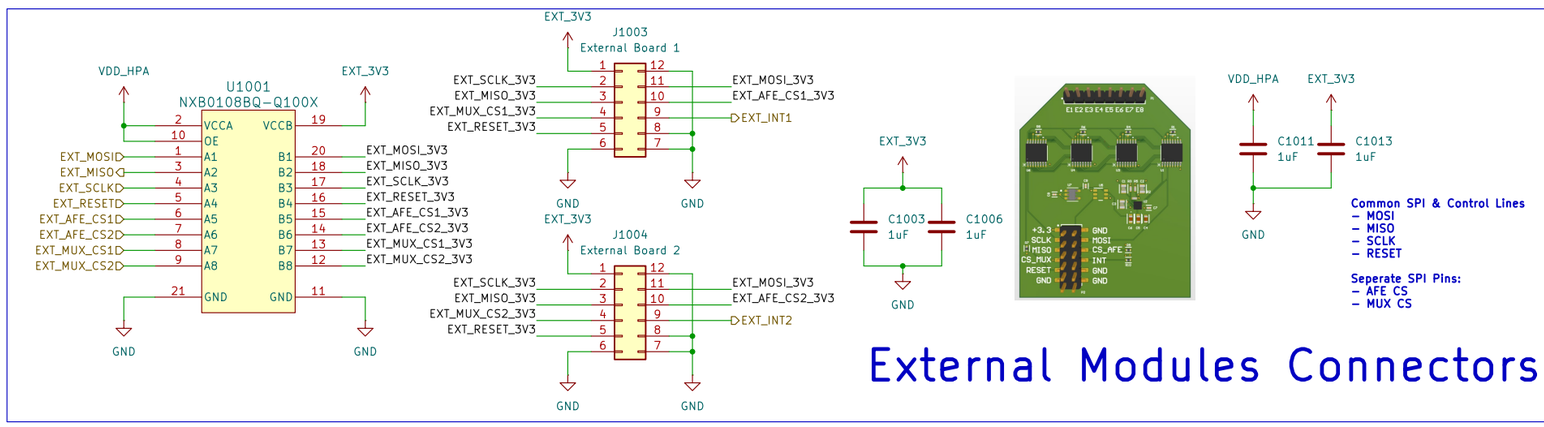

1V8D
C806 100nF
C810 100nF
GND
3V3D
C803 100nF
C807 100nF
C811 100nF
C814 100nF
GND
1V8A
C804 100nF
C808 100nF
C812 100nF
GND
3V3A
C801 100nF
C802 100nF
C805 100nF
C809 100nF
C813 100nF
C815 100nF
C816 100nF
GND
U801
AD8285WBCPZ
Forces the auxiliary channel (INADC) to the ADC.
AFE_AUX
C818 100nF
R801 10K
GND
CLK_N
R803 100R
C819 100nF
CLK_P
AFE_CSn
AFE_CLK_AC_N
AFE_CLK_AC_P
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
3V3D
1V8D
Forces channel A to the ADC if AUX is not asserted.
AFE_MUXA
R802 10K
GND
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R805
R804 10K
R_series recommended by datasheet
GND
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AFE_VREF
C817 1uF
C820 100nF
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8A
3V3A
3V3D
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C823 100nF
C824 100nF
C825 100nF
C826 100nF
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
D801 MMBD4148SE
D802 MMBD4148SE
D803 MMBD4148SE
D804 MMBD4148SE
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R806 8R
R807 8R
R808 8R
R809 8R
Change value according to pulser T/R switch out impedance
C821 100nF
C822 100nF
C827 100nF
C828 100nF
GND

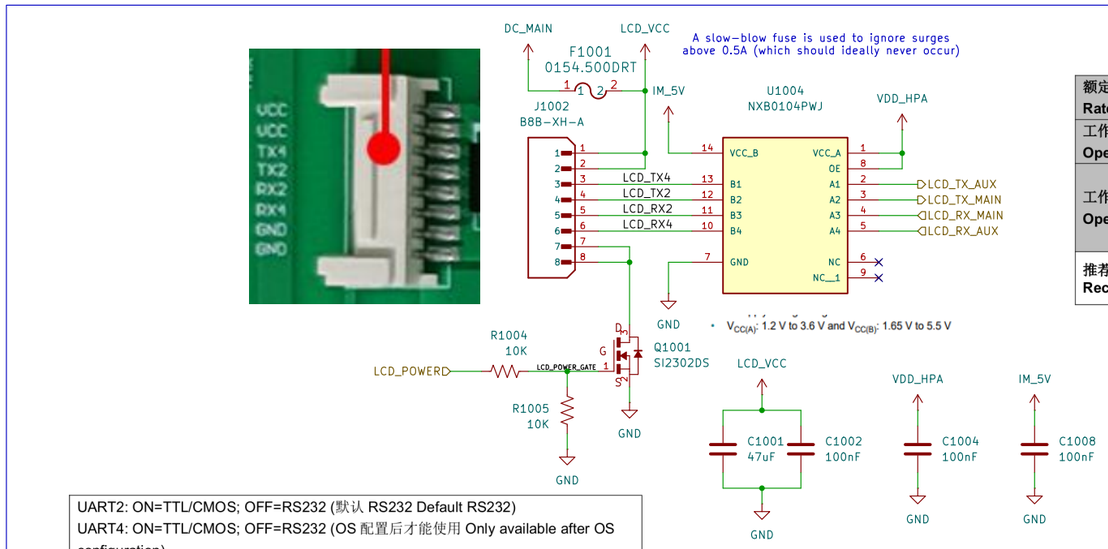

1V8D
C906 100nF
C910 100nF
GND
3V3D
C903 100nF
C907 100nF
C911 100nF
C914 100nF
GND
1V8A
C904 100nF
C908 100nF
C912 100nF
GND
3V3A
C901 100nF
C902 100nF
C905 100nF
C909 100nF
C913 100nF
C915 100nF
C916 100nF
GND
U901
AD8285WBCPZ
Forces the auxiliary channel (INADC) to the ADC.
AFE_AUX
R901 10K
GND
C918 100nF
CLK_N
R903 100R
C919 100nF
CLK_P
AFE_CSn
AFE_CLK_AC_N
AFE_CLK_AC_P
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
3V3D
1V8D
Forces channel A to the ADC if AUX is not asserted.
AFE_MUXA
R902 10K
GND
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R905
R904 10K
R_series recommended by datasheet
GND
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AFE_VREF
C917 1uF
C920 100nF
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8A
3V3A
3V3D
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C923 100nF
C924 100nF
C925 100nF
C926 100nF
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
D901 MMBD4148SE
D902 MMBD4148SE
D903 MMBD4148SE
D904 MMBD4148SE
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R906 8R
R907 8R
R908 8R
R909 8R
Change value according to pulser T/R switch out impedance
C921 100nF
C922 100nF
C927 100nF
C928 100nF
GND



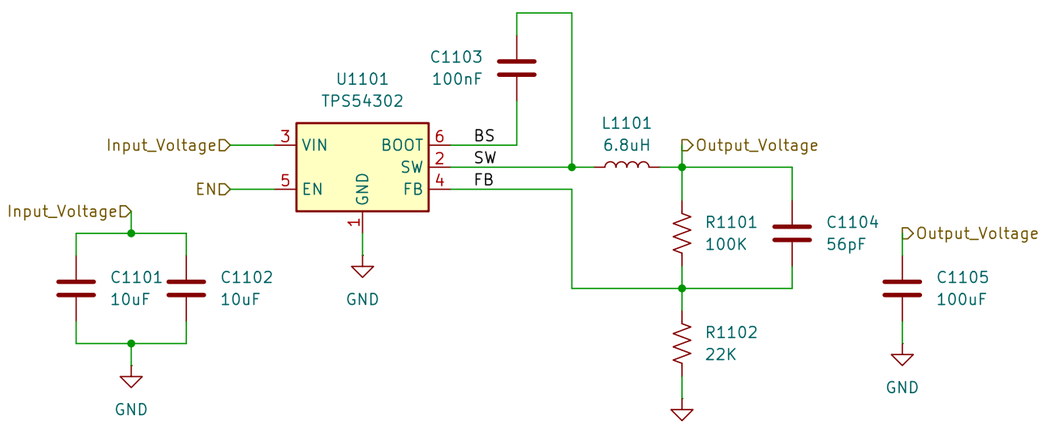
CFG	14	D	I	V_{DDIO}	REFCLK clock frequency configuration pin. Two frequencies are supported: 19.2 MHz when 0, or 26 MHz when 1.
CS	11	D	I	V_{DDIO}	Active-high chip select pin. When low the IC is in power down and ULPI bus is tri-stated. When high normal operation. Tie to V_{DDIO} if unused.



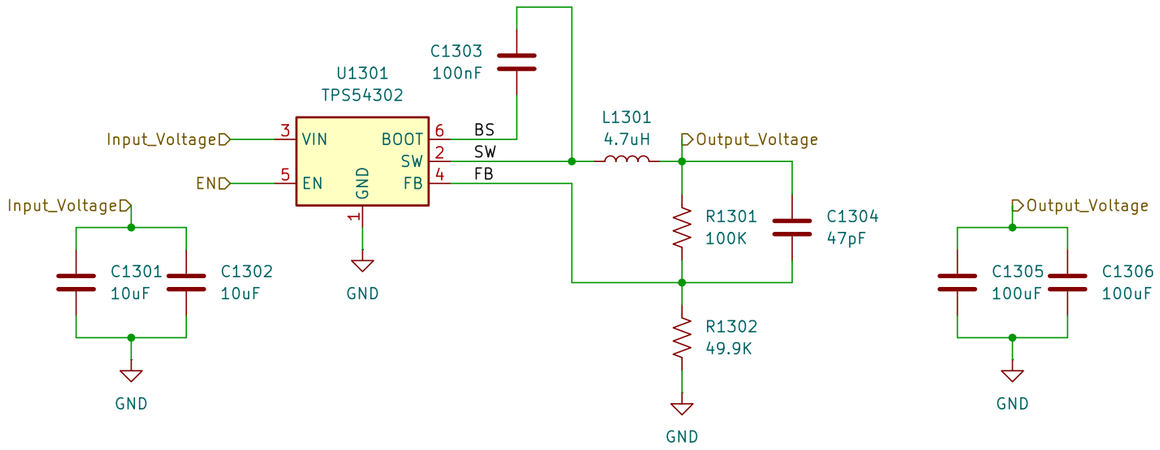




额定功率 **Rated Power**	<5W	
工作电压 **Operating Voltage**	9~36V，典型值 12V 9~36V, typical value of 12V	
工作电流 **Operating Current**	320mA	VCC=12V，背光亮度最大 VCC=12V, max backlight
	120mA	VCC=12V，背光关闭 VCC=12V, backlight off
推荐工作电源：12V 1A 的直流稳压电源 **Recommended power supply: 12V 1A DC**		



WEBENCH DESIGNER LINK (3.3V)



WEBENCH DESIGNER LINK (1.8V)

Duty Cycle

$$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$$ = (100+0.7) / (11+100+0.7) = 90%

= (50+0.7) / (11+50+0.7) = 82%

F_switching = 150 KHz

Inductor

$$L1 = L2 = L3 = \frac{V_{inmin} \times V_{inmax} \times D_{max}}{2 \times I_{out} \times V_{out} \times 0.4 \times f_{SW}}$$ = (11*25*0.9) / (2*0.3*100*0.4*150000) = 68uH

n = 0.88

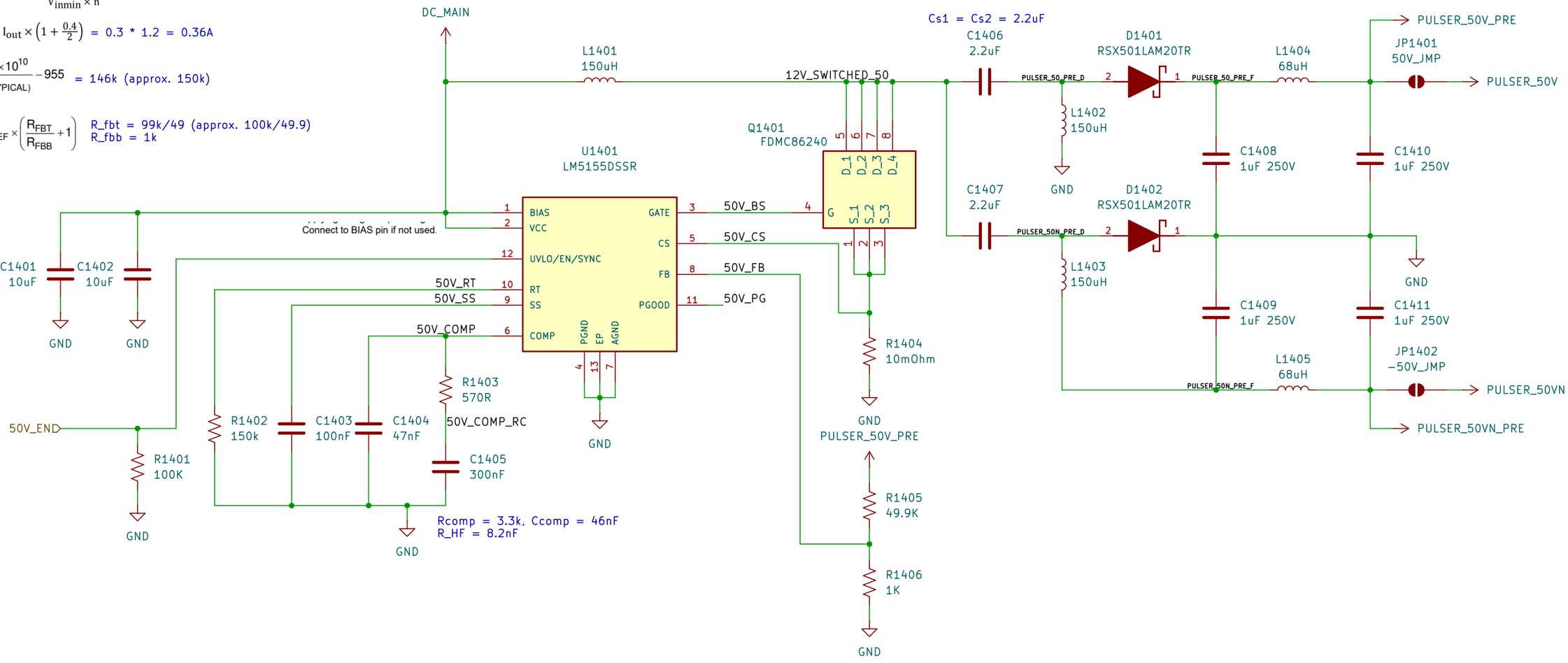
$$I(L1_{peak}) = \frac{I_{out} \times (V_{out} + V_d) \times \left(1 + \frac{0.4}{2}\right)}{V_{inmin} \times n}$$ = (0.3*100.7*1.2) / (11*0.88) = 3.75A

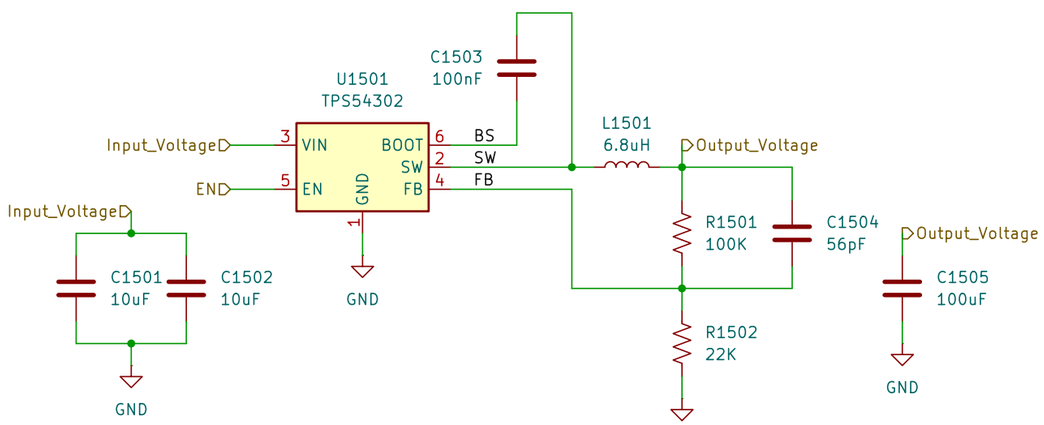
$$I(L2_{peak}) = I_{out} \times \left(1 + \frac{0.4}{2}\right)$$ = 0.3 * 1.2 = 0.36A

$$R_T = \frac{2.21 \times 10^{10}}{f_{RT(TYPICAL)}} - 955$$ = 146k (approx. 150k)

$$V_{LOAD} = V_{REF} \times \left(\frac{R_{FBT}}{R_{FBB}} + 1\right)$$ R_fbt = 99k/49 (approx. 100k/49.9)

R_fbb = 1k





WEBENCH DESIGNER LINK (3.3V)

Duty Cycle

$$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$$ = (100+0.7) / (11+100+0.7) = 90%

= (50+0.7) / (11+50+0.7) = 82%

F_switching = 150 KHz

Inductor

$$L1 = L2 = L3 = \frac{V_{inmin} \times V_{inmax} \times D_{max}}{2 \times I_{out} \times V_{out} \times 0.4 \times f_{SW}}$$ = (11*25*0.9) / (2*0.3*100*0.4*150000) = 68uH

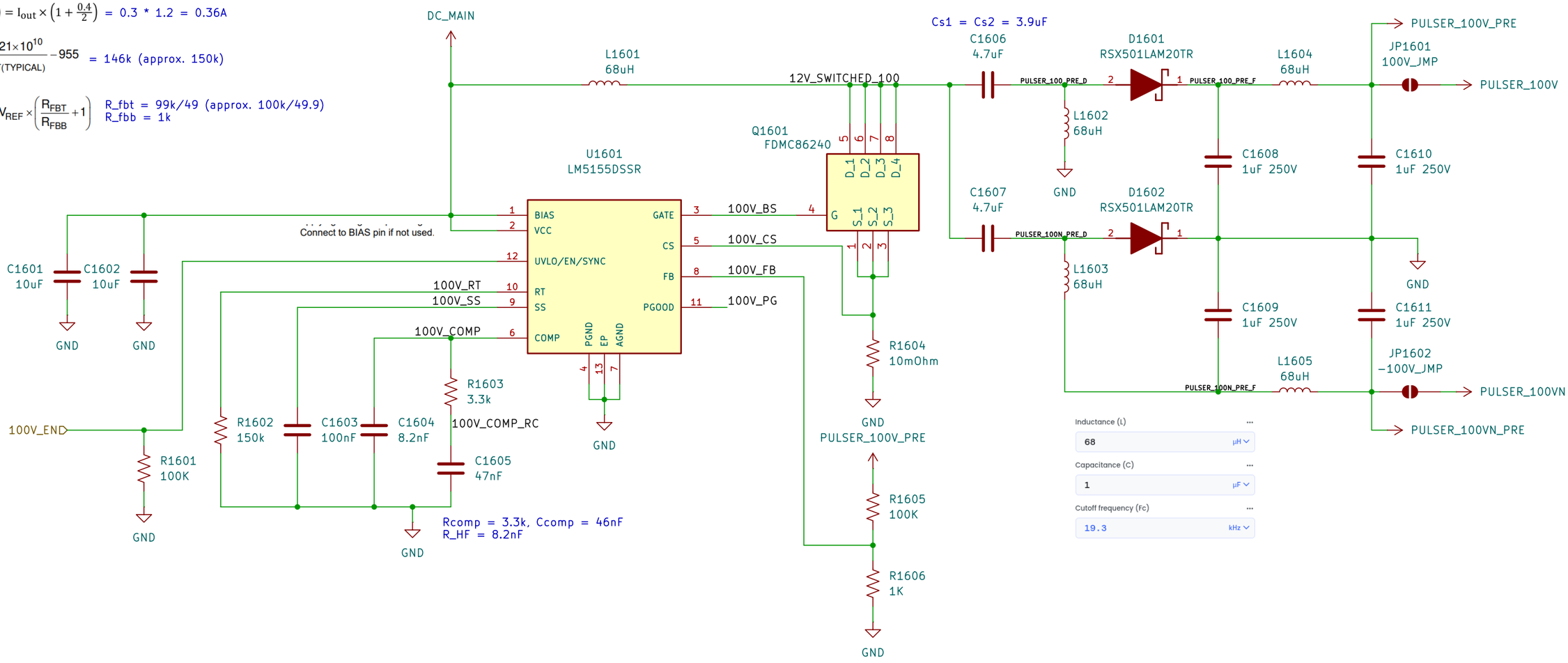
n = 0.88

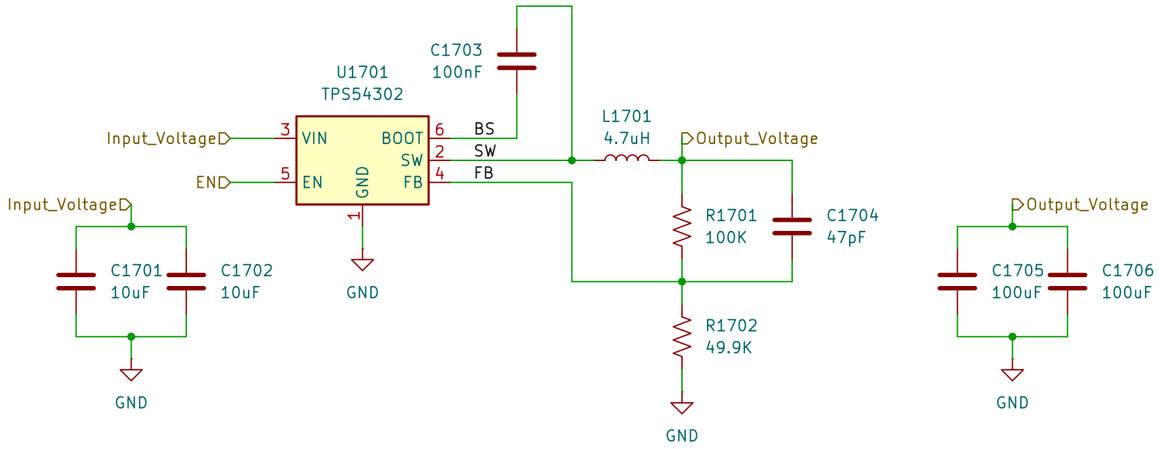
$$I(L1_{peak}) = \frac{I_{out} \times (V_{out} + V_d) \times \left(1 + \frac{0.4}{2}\right)}{V_{inmin} \times n}$$ = (0.3*100.7*1.2) / (11*0.88) = 3.75A

$$I(L2_{peak}) = I_{out} \times \left(1 + \frac{0.4}{2}\right)$$ = 0.3 * 1.2 = 0.36A

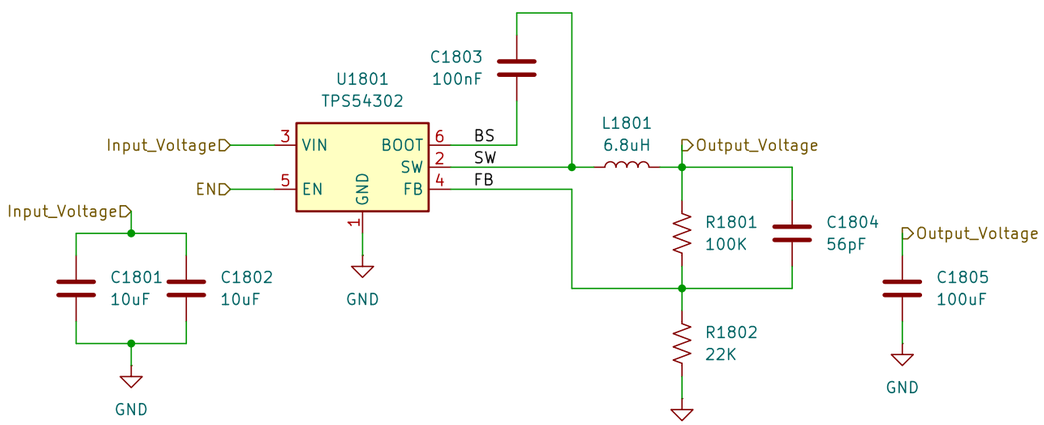
$$R_T = \frac{2.21 \times 10^{10}}{f_{RT(TYPICAL)}} - 955$$ = 146k (approx. 150k)

$$V_{LOAD} = V_{REF} \times \left(\frac{R_{FBT}}{R_{FBB}} + 1\right)$$ R_fbt = 99k/49 (approx. 100k/49.9)
R_fbb = 1k

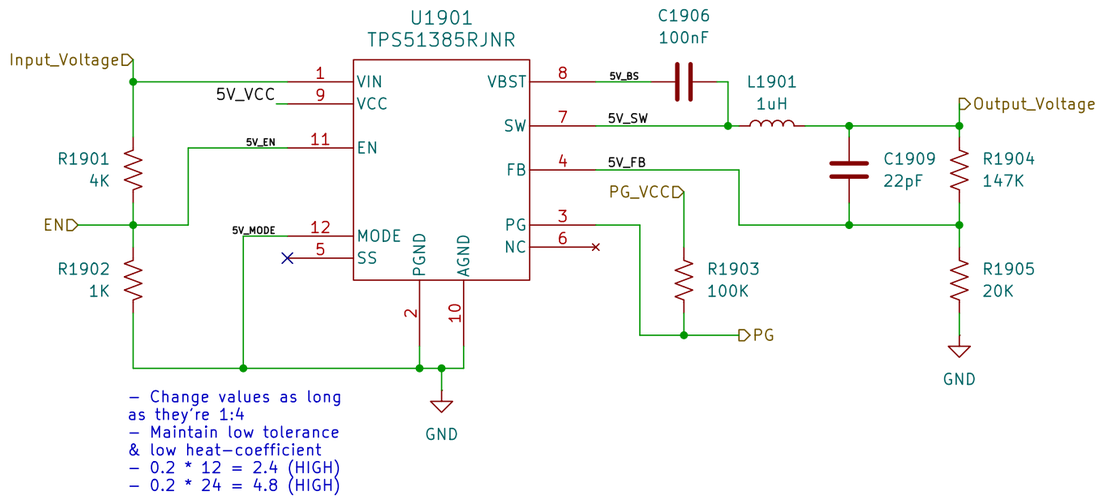


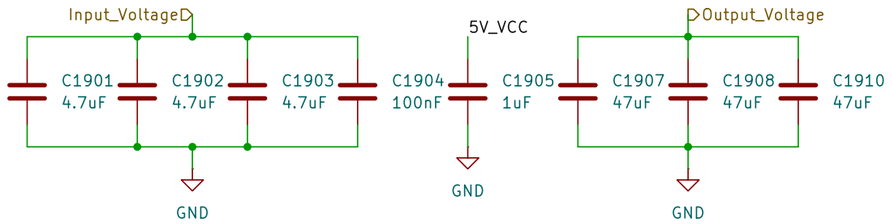


WEBENCH DESIGNER LINK (1.8V)

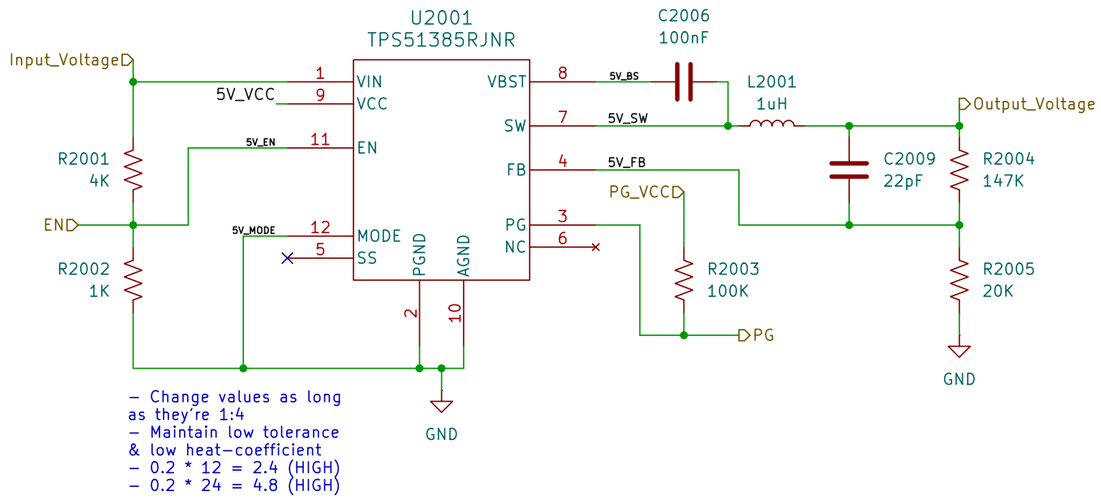


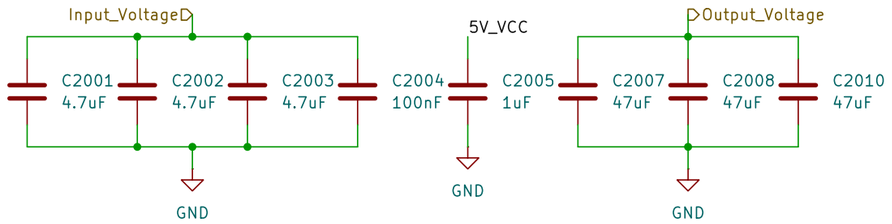
WEBENCH DESIGNER LINK (3.3V)





WeBench Designer Link





WeBench Designer Link